

ASSIGNMENT #1

Course: CHE 341. CHEMICAL PROCESS CONTROL
Term: Spring 2020
Professor: Dr. Michael Caracotsios
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Due Date: Wednesday January 22 by 5:00pm in my office

Number of points.....100

Your score.....

87
93
96

Note: All problems are worth 100 points. You final score will be the average value

2. Calculations/Tables

Table II: Calculated parameters from ASPEN simulation of CSTR.

Parameters	Feed	Effluent
Density, ρ [kmol/m ³]	17.8 ✓	✓ 16.7
Heat Capacity, C_p [kJ/kmol·K]	99.96 ✓	✓ 133.1
Heat of Reaction, ΔH°_R [kJ/kmol]	-29070 ✓	
Equilibrium Constant, K_{eq}	60.78 ✓	
Water flow rate, $\dot{m}$ [kmol/hr]	974.840 ✓	✓
Overall heat transfer coefficient, U [kJ/m ² ·min·K]	659.5 ✓	

Heat of Reaction

All values for enthalpy, H , were obtained in ASPEN. See Table I.

$$(1) \quad \Delta H_R = \sum n_i H(\text{products}) - \sum n_i H(\text{reactants})$$

$$\Delta H_R = [H(EAC) + H(W)] - [H(AA) + H(EtOH)]$$

$$= [(-471830 - 282380)] - [(-453510 - 271630)] \frac{\text{kJ}}{\text{mol}} = -29070 \frac{\text{kJ}}{\text{kmol}}$$

Equilibrium Constant, K_{eq}

$$(2) \quad K = \prod (x_i \gamma_i)^{v_i} = (0.62 \cdot 0.18)^{-1} (1.34 \cdot 0.17)^{-1} (1.69 \cdot 0.32) (2.07 \cdot 0.34) = 15.18$$

$$(3) \quad K_{eq} = \exp\left(-\frac{\Delta G^\circ}{RT}\right) = \exp\left(-\frac{(-74398.1 - 54945.1) - (-88589.5 - 37952.9) \frac{\text{cal}}{\text{mol}} \left(\frac{4.184 \text{ J}}{\text{cal}}\right)}{8.3145 \frac{\text{J}}{\text{mol} \cdot \text{K}} (343.15 \text{ K})}\right) = 60.78$$

The Gibb's Free Energy, liquid activity coefficients, and mole fractions were obtained from the effluent stream of the ASPEN simulation (see Table I). When $K_{eq} = K$, then the reaction is in equilibrium. In this case, $K_{eq} > K$, therefore the reaction is not at equilibrium.

Water Flow Rate, $\dot{m}$

$$(4) \quad Q = \dot{m} C_p \Delta T \Rightarrow \dot{m} = \frac{Q}{C_p \Delta T} = \frac{-81240 \frac{\text{cal}}{\text{s}} \left(\frac{3600 \text{ s}}{\text{hr}}\right) \left(\frac{4.184 \text{ J}}{\text{cal}}\right)}{75.3 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}} \left(\frac{1000 \text{ J}}{\text{kJ}}\right) (305.35 \text{ K} - 322.04 \text{ K})} = 974.840 \frac{\text{kmol}}{\text{hr}}$$

Overall Heat Transfer Coefficient, U

First, we need to find the area, A , by substituting $L = 2D$ in the volume equation:

$$V = 500 \text{ L} \left(\frac{1 \text{ m}^3}{1000 \text{ L}}\right) = \frac{\pi D^2}{4} * 2D \Rightarrow D = 0.6828 \text{ m} \Rightarrow A = \pi D^2 = 1.465 \text{ m}^2$$

$$(5) \quad Q = UA(T_{cw} - T_r) \Rightarrow U = \frac{Q}{A(T_{cw} - T_r)} = \frac{-81240 \frac{\text{cal}}{\text{s}} \left(\frac{60 \text{ s}}{\text{min}}\right) \left(\frac{4.184 \text{ J}}{\text{cal}}\right) \left(\frac{1 \text{ kJ}}{1000 \text{ J}}\right)}{1.465 \text{ m}^2 (322.04 \text{ K} - 343.15 \text{ K})} = 659.5 \frac{\text{kJ}}{\text{m}^2 \cdot \text{min} \cdot \text{K}}$$

4. Athena Code

!Variable Declarations and Data

!*****

Parameter Rg=8.3145 As Real !Universal Gas Constant, J/(mol K)

Global Q, FAAfeed, FEtfeed, FEACfeed, FWfeed, Tfeed As Real !Feed Stream variables

Q = 0.365 !Volumetric flow rate, m³/min

FAAfeed = 193.0/60 !Flow rate of Acetic Acid in feed, kmol/min

FEtfeed = 187.0/60 !Flow rate of Ethanol in feed, kmol/min

FEACfeed = 0.0 !Flow rate of Ethyl Acetate in feed, kmol/min

FWfeed = 10.0/60 !Flow rate of Water in feed, kmol/min

Tfeed = 25 !Temperature of feed, C

Global Vol, UA, Tcw As Real

!Reactor spec variables

Vol = 0.50 !Volume of Reactor, m³

UA = 966.0 !Heat Transfer Coefficient*Area, kJ/(K*min)

Tcw = 50 !Temperature of cooling water, C

Global rho, Cp, DHR1 As Real

!Thermodynamic variables

rho = 17.8 !Feed Density of the mixture, kmol/m³

Cp = 100 !Heat capacity of mixture, kJ/(kmol*K)

DHR1 = -29070 !Heat of reaction, kJ/kmol

Global kfo, kro, Er As Real

!Kinetics variables

kfo = 1.9e8 !Pre-exponential factor for forward rxn, m³/(kmol*s)

kro = 5.0e7 !Pre-exponential factor for reverse rxn, m³/(kmol*s)

Er = 7156.2 !Activation energy for forward and reverse rxns, K

@Initial Conditions

U(1) = 0.50*rho

U(2) = 0.0

U(3) = 0.0

U(4) = 0.50*rho

U(5) = 50

@Model Equations

Dim CAA, CEt, CEAC, CW, Temp As Real

CAA = U(1) !Concentration of acetic acid, kmol/m³

CEt = U(2) !Concentration of ethanol, kmol/m³

CEAC = U(3) !Concentration of ethyl acetate, kmol/m³

CW = U(4) !Concentration of water, kmol/m³

Temp = U(5) + 273.15 !Temp of reactor, K

Dim kf, kr, R1 As Real

kf = kfo*exp(-Er/Temp) !Forward rate constant, m³/(kmol*s)

kr = kro*exp(-Er/Temp) !Reverse rate constant, m³/(kmol*s)

R1 = (kf*CAA*CEt) - (kr*CEAC*CW) !Reaction Rate, m³/(kmol*s)